COPYRIGHT RESERVED BCA (Sem-III) – Mathematics

(305) GE-3

2025

Time : 1:30 hours

Full Marks : 40

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer all group as directed.

Group—A

(Objective Type Questions)

1. Choose the correct answer : $1 \times 10 = 10$

(i) In tossing three coins at a time, the probability of getting at most one head is:

(a) $\frac{3}{8}$ (b) $\frac{7}{8}$

(c) $\frac{1}{2}$ (d) None of these

(ii) The probability that a leap year selected at random will be 53 Sundays is :

(a) $\frac{3}{7}$ (b) $\frac{1}{7}$

(c) $\frac{2}{7}$ (d) None of these

(iii) The probability of an impossible event is :

(a) 0 (b) 1

(c) $\frac{1}{2}$ (d) None of these

(iv) The Probability can take value :

(a) $-\infty$ $+\infty$ (b) 1 to ∞

(c) 0 to 1 (d) None of these

(v) Mean is a measure of :

(a) Location (b) Dispersion

(c) Correlation (d) None of these

(vi) The correct relation between AM, GM and HM is :

(a) AM = GM = HM (b) AM > GM > HM

(c) GM > AM > HM (d) None of these

(vii) The term regression was introduced by :

(a) R.A. Fisher (b) Sir Francis Galton

(c) Karl Pearson (d) None of these

(viii) The Arithmetic mean of 15, 20, 25, 12 is :

(a) 18 (b) 16

(c) 17 (d) None of these

(ix) The mode of the samples 2, 1, 3, 2, 5, 3, 2, 4, 2, 1, 6 is :

(a) 1 (b) 2

(c) 3 (d) None of these

(x) The value of correlation ratio varies from:

(a) 0 to 1 (b) -1 to 0

(c) -1 to 1 (d) None of these

Group – B

(Short Answer Type Questions)

Answer any **three** of the following : $5 \times 3 = 15$

2. What do you understand by measure of central tendency?

3. State addition and multiplication theorem of probability.

4. Define arithmetic mean with properties.

5. Define correlation coefficient.

6. Discuss about binomial distribution.

7. How will you calculate median in case of ungrouped data?

Group – C

(Long Answer Type Questions)

Answer any **two** of the following : 7.5×2=15

8. Define corelation co-effiicient and how it differ from regression.

9. Explain Bayes' theorem with an example.

10. Calculate the mean from the following data:

Class-interval	10-14	14-18	18-22	22-26	26-30	30-34	34-38	38-42
Frequency	6	7	8	10	9	3	3	1

11. 4 cards are drawn from a well-shuffled deck of 52. What is the probability of obtaining 3 diamonds and one spade?

———×———